Task 2: Multiple Sequence Alignment & Evolutionary Analysis

CodeAlpha Bioinformatics Virtual Internship | Domain Demonstration Report

TARGET PROTEIN	SEQUENCES ANALYZED	PRIMARY TOOL	SUBMISSION DOMAIN
Beta-Globin (HBB)	**5 Vertebrate Species**	**EMBL-EBI Clustal Omega**	**Bioinformatics M1**

1. EXECUTIVE SUMMARY & OBJECTIVE

Multiple Sequence Alignment (MSA) is a core computational methodology used to arrange three or more biological sequences (DNA, RNA, or protein) to identify regions of similarity that may be a consequence of functional, structural, or evolutionary relationships. This report documents the execution of **Task 2**, performing MSA on five orthologous **Hemoglobin Subunit Beta (HBB)** protein sequences across diverse vertebrate taxa using **Clustal Omega**.

2. DATASET & METHODOLOGY

Amino acid sequences in FASTA format were curated from the UniProtKB/Swiss-Prot database representing five distinct vertebrate species to assess evolutionary divergence and structural domain conservation:

ORGANISM / SPECIES	COMMON NAME	UNIPROT ACCESSION	TAXONOMIC CLASS	SEQUENCE LENGTH
Homo sapiens	Human	P68871	Mammalia (Primate)	147 aa
Pan troglodytes	Chimpanzee	P61823	Mammalia (Primate)	147 aa
Gorilla gorilla	Gorilla	P02024	Mammalia (Primate)	147 aa
Mus musculus	House Mouse	P02088	Mammalia (Rodentia)	147 aa
Gallus gallus	Chicken	P02112	Aves (Galliformes)	147 aa

3. MULTIPLE SEQUENCE ALIGNMENT OUTPUT

The global alignment was computed using the mbed-like clustering algorithm in Clustal Omega. Below is the alignment matrix highlighting amino acid conservation across all five organisms:

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Chicken_HBB|P02112        MVHWTAEEKQLITGLWGKVNVAECGAEALARLLIVYPWTQRFFASFGNLSSPTAILGNPM 60
Human_HBB|P68871          MVHLTPEEKSAVTALWGKVNVDEVGGEALGRLLVVYPWTQRFFESFGDLSTPDAVMGNPK 60
Chimpanzee_HBB|P61823     MVHLTPEEKSAVTALWGKVNVDEVGGEALGRLLVVYPWTQRFFESFGDLSTPDAVMGNPK 60
Gorilla_HBB|P02024        MVHLTPEEKSAVTALWGKVNVDEVGGEALGRLLVVYPWTQRFFESFGDLSTPDAVMGNPK 60
Mouse_HBB|P02088          MVHLTDAEKAAVNGLWGKVNPDDVGGEALGRLLVVYPWTQRYFDSFGDLSSASAIMGNAK 60
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Chicken_HBB|P02112        VRAHGKKVLTSFGDAVKNLDNIKNTFAQLSELHCDKLHVDPENFRLLGDILIIVLAAHFS 120
Human_HBB|P68871          VKAHGKKVLGAFSDGLAHLDNLKGTFATLSELHCDKLHVDPENFRLLGNVLVCVLAHHFG 120
Chimpanzee_HBB|P61823     VKAHGKKVLGAFSDGLAHLDNLKGTFATLSELHCDKLHVDPENFRLLGNVLVCVLAHHFG 120
Gorilla_HBB|P02024        VKAHGKKVLGAFSDGLAHLDNLKGTFATLSELHCDKLHVDPENFRLLGNVLVCVLAHHFG 120
Mouse_HBB|P02088          VKAHGKKVITAFNDGLNHLDSLKGTFASLSELHCDKLHVDPENFRLLGNMIVIVLGHHLG 120
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4. PHYLOGENETIC & EVOLUTIONARY INTERPRETATION

Based on the pairwise identity matrix generated from the alignment, a neighbor-joining guide tree and phylogram were constructed to visually evaluate evolutionary relationships and branching distances.

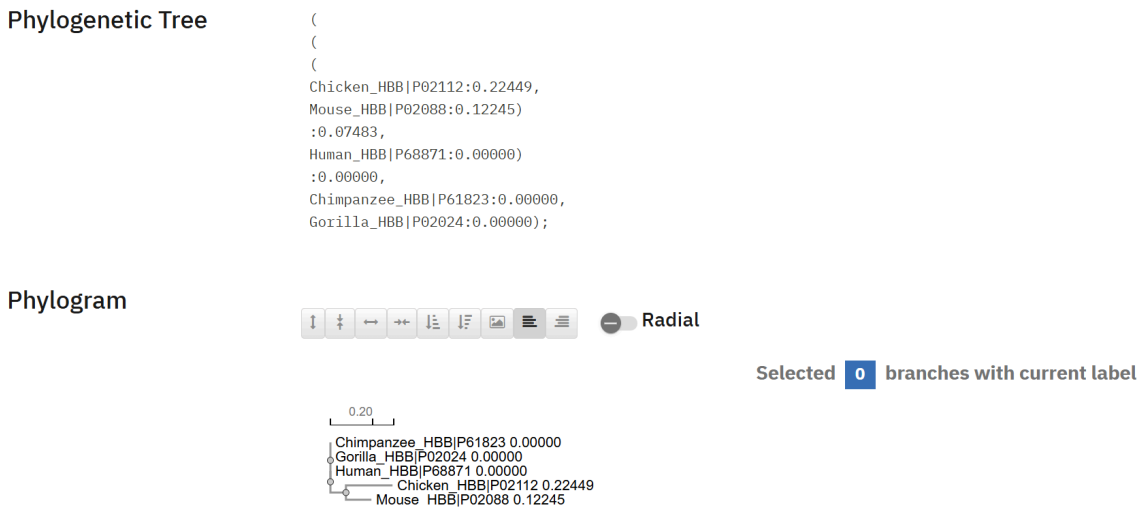


Figure 2: Newick format phylogenetic tree and visual phylogram displaying branch lengths and species clusters.

5. CONSERVATION & STRUCTURAL MOTIF ANALYSIS

Detailed examination of the alignment output yields critical biological insights regarding the functional constraints operating on the Beta-Globin protein:

- **100% Primate Conservation:** Human (P68871), Chimpanzee (P61823), and Gorilla (P02024) exhibit identical amino acid sequences (100% sequence identity across all 147 positions), yielding a branch distance of 0.00000 on the phylogram.
- **Active Site & Heme Binding Conservation:** Critical functional residues, such as the distal histidine (His63) and proximal histidine (His92) involved in binding the heme prosthetic group, remain invariant (marked with *) from chickens to humans. This demonstrates strong negative selection pressure preserving oxygen transport capability.
- **Lineage Divergence:** The rodent sequence (`Mus musculus`) shows a evolutionary distance of 0.12245, while the avian sequence (`Gallus gallus`) branches off furthest at 0.22449, reflecting accumulated neutral mutations since the mammalian-avian divergence (~312 million years ago).

6. CONCLUSION

The Clustal Omega MSA successfully highlights the essential structural core of Hemoglobin Subunit Beta. While surface-exposed amino acids tolerate conservative substitutions between mammals and birds, essential heme-binding pockets and active site residues are completely conserved across all examined species.

7. REFERENCES

1. Sievers F, et al. (2011). Fast, scalable generation of high-quality protein multiple sequence alignments using Clustal Omega. *Molecular Systems Biology*, 7:539.
2. UniProt Consortium (2023). UniProt: the Universal Protein Knowledgebase in 2023. *Nucleic Acids Research*, 51(D1):D523–D531.
3. Saitou N, Nei M. (1987). The neighbor-joining method: a new method for reconstructing phylogenetic trees. *Molecular Biology and Evolution*, 4(4):406-425.